Information Architecture of the TERRA Studio Mode: Evaluating a Land-Cover Model Across Domains

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Abstract

The TERRA studio is a WebGL surface that detaches analysis rasters from geographic coordinates so that several areas of interest (AOIs) can be read together. This document records why that surface exists, what scientific question it is built to answer, and the constraint that follows from the answer. The question is model transferability: given a classifier fitted on one region, how does its accuracy behave on regions it was not fitted on, and is the degradation attributable to distributional shift. That question imposes a *star* topology — one privileged reference domain against N target domains — rather than the pairwise topology the interface was first built for, and the distinction is not cosmetic: pairwise devices scale as $\binom{N}{2}$ while the analysis itself scales as N. A static audit of the result payload found approximately 822 transitively declared fields, of which the studio surfaces on the order of 60 to 70; three quantities computed by the sidecar are dropped in transit and cannot reach any surface, including the per-feature table that localises distributional distance to named features. Two constraints bound any response and are measured here: the application's minimum window of 1000×700 pixels, at which existing chrome already consumes half the width, and the fact that the editors a first reading would propose building already exist as separate screens. Blender is examined as a reference architecture for dense professional interfaces, with mechanisms separated into those that transfer and those that do not, and supplemented from scientific visualisation where Blender is silent — notably on what makes two measured rasters comparable at all. Limitations are stated in Section 10.

1 Scope and purpose of this document

The studio was built incrementally, and its surfaces — an outliner column, a selection column, a detail band, a run band, two modal dialogs — were each added to answer a local need. The code records what was built. It does not record which scientific question the arrangement serves, and without that record the arrangement cannot be evaluated, only adjusted.

This document states the question, derives the information structure it implies, measures the gap between that structure and the current interface, and describes a reference architecture. It is a design record, not an implementation plan.

2 The scientific premise

2.1 The evaluation scenario

A land-cover classifier is fitted on labelled samples drawn from one region. The operational question is whether it can be applied elsewhere. In TERRA this is asked concretely: an AOI is imported on the map, a run is executed, and the result is brought into the studio. Repeating this for regions that differ from the training region — in phenology, soil background, cloud regime, crop calendar and sensor viewing geometry — produces a set of results that are comparable in structure and different in provenance.

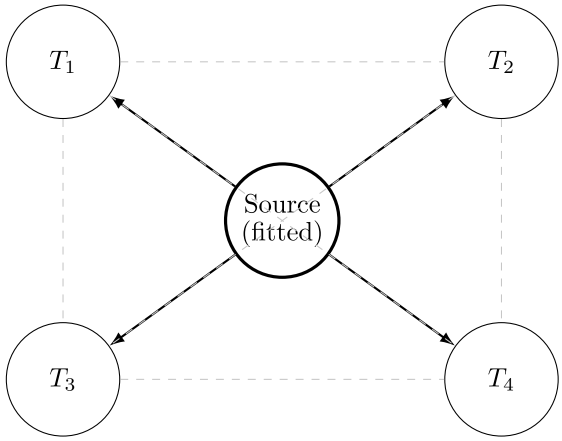

Figure 1: Comparison topology for $N = 5$. Solid arrows are the $N - 1$ comparisons the transferability question defines; dashed edges are the remaining $\binom{N}{2} - (N - 1)$ pairs, which are computable and do not bear on it. A pairwise comparison device does not distinguish between the two kinds.

A representative case: a model fitted in the Brazilian Northeast, then applied in the South, the North and the West. Each target region carries different spectral and phenological characteristics. The number of AOIs is not bounded by the method; it is bounded by how many regions the analyst chooses to test.

2.2 The topology is a star, not a graph

The set of AOIs is not symmetric. One of them — the training region — is privileged, and every other is measured against it. Comparisons among the target regions are defined but do not answer the question being asked.

For N AOIs this is the difference between $N - 1$ meaningful comparisons and $\binom{N}{2}$ possible ones. With $N = 6$: five comparisons carry the analysis, and fifteen pairs exist. An interface whose primary comparison device is pairwise therefore presents ten arrangements that answer nothing, and requires $N - 1$ separate invocations to assemble the answer it does support.

2.3 Why the shift is not covariate shift

Land cover is anticausal in the sense of Schölkopf et al. (2012): the class generates the observed reflectance, not the reverse. Covariate shift assumes $P(Y \mid X)$ is stable while $P(X)$ moves, which is the causal direction and does not hold here. What moves between regions is the class prior $P(Y)$ — crop composition differs — together with the class-conditional appearance $P(X \mid Y)$, since the same crop presents differently under a different soil background and calendar. This combination is generalized target shift in the sense of Zhang et al. (2013), and it is the regime the studio's diagnostics must describe.

2.4 The figure the question resolves to

Ben-David et al. (2010) bound the target error of a hypothesis h by

$$\varepsilon_T(h) \;\leq\; \varepsilon_S(h) \;+\; d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T) \;+\; \lambda, \tag{1}$$

where ε_S is the source error, $d_{\mathcal{H}\Delta\mathcal{H}}$ a divergence between the two marginals, and λ the error of the best joint hypothesis. The bound states that divergence constrains the error gap. It does not state that the constraint is tight, and whether it is tight for a given model and sensor is an empirical question.

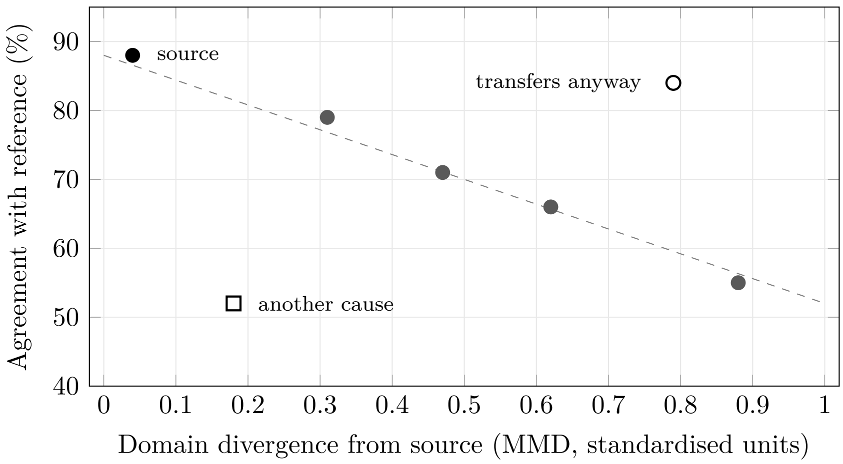

Figure 2: Divergence against accuracy, one point per AOI. Illustrative values, not measurements. The arrangement is the analysis rather than a summary of it: the trend tests whether Eq. (1) is tight for this model, and the two annotated outliers are the cases worth investigating. The figure scales linearly in the number of AOIs.

That question is answered by plotting divergence against observed accuracy for all N AOIs at once (Figure 2). Each AOI contributes one point. Three readings follow directly from the arrangement:

- Points falling near a descending trend indicate that divergence accounts for the observed degradation, and the bound in Eq. (1) is informative for this model.
- A point at large divergence and high accuracy indicates transfer despite distance, which suggests the divergence measure is capturing variation the classifier is invariant to.
- A point at small divergence and low accuracy indicates that the cause is elsewhere — label definitions, reference quality, spatial resolution, or the λ term — and not distributional distance.

This is one figure with N points, and it is the deliverable of the transferability study. The studio does not currently produce it.

3 Measured pressure on the interface

3.1 Field count, and what it does and does not mean

A static traversal of `PredictResult` and its transitive members in `frontend/src/lib/types.ts` reaches approximately 822 declared fields. The studio reads on the order of 60 to 70 of them. Three whole product payloads, accounting for roughly 576 fields, have no studio surface.

This figure is easy to misread and the qualification matters. Those payloads are *not* unrendered by the application: `pages/AnalysisPage.tsx` mounts `WindScreening`, `EnergyModelSection`, the solar resource, terrain and siting sections and `LulcSection`, each gated on the presence of its payload rather than greyed out when absent — which is already the discipline described in Section 5.4. The application renders them; the studio does not.

The number therefore measures one thing only: how much of a result is outside the studio's reach, and so how much a multi-AOI comparison performed there cannot consult. It is not evidence of a missing feature elsewhere.

3.2 Surfaces that already exist

A design record that proposes building what is already built is worse than no record. Four components bear directly on the proposals in Section 6 and are present in the codebase:

- `frontend/src/components/DataTableView.tsx` renders a sortable table with missing-value-last ordering, clipboard copy and CSV download.
- `frontend/src/lib/analysisTables.ts` exports on the order of 28 table builders keyed on payload presence, among them `domainFingerprintTable`, `lulcPredVsRefTable` and the wind and energy families.
- `ResearchPackModal` lists every table a result carries with its row count, named after the files the exporter writes, so an inspection maps one-to-one onto the exported pack.
- `frontend/src/components/whiteboard/boardMemory.ts` is a module-scoped keyed record that outlives component remounts, holding per-slot state across a visit to another screen.

The tabular capability described in Section 6.2 therefore exists as a *component* and is absent as a *studio surface*. What is missing is the routing, not the table.

`boardMemory` additionally records a deliberate position that one proposal below contradicts: its documentation states that it is not persistence, that it survives a close and not a restart, and that saving is something a user asks for by name. Any proposal to persist disclosure state is a reversal of that decision and must be argued against it rather than presented as filling a gap.

3.3 Quantities computed and then dropped

Three outputs of `sidecar/domain_shift.py` are not carried through the Go type that declares the report, and therefore cannot reach the interface at all. Each is load-bearing for the interpretation of the others.

- `feature_shift` — a per-feature table giving, for the twelve features with the largest displacement, the standardised means in each domain, the gap in training standard deviations, and the classifier's impurity importance. A scalar divergence states that two domains differ; this table states *where*. Without it a large distance is not actionable.
- `same_space` and `standardised` — qualifiers stating whether two fingerprints are comparable at all. Comparing an 80-feature spectral fingerprint against a single-column NDVI fingerprint produces a number from two different quantities. Without these flags every distance shown is unqualified.
- `cva_magnitude_sd` — change-vector magnitude expressed in training standard deviations. The interface currently shows the raw magnitude. Measured on the shipped artifacts, six acquisition-index features account for 99.7% of the raw squared distance while the sixteen reflectance features account for the remainder, so the raw figure is dominated by quantities that carry no domain meaning. Standardising moved one reflectance shift from a factor of 1.4 to 5.5.

3.4 The geometric budget

Any proposal that adds or widens a surface is bounded by a figure that is set in the application and not negotiable at the level of a stylesheet: `app.go` sets a minimum window of 1000×700 pixels.

At that minimum the two columns consume $240 + 255 = 495$ px, which is 49.5 % of the width, and the two foot bands consume 212 px of 700, which is 30.3 %. The canvas the studio exists to present therefore receives approximately 505×488 px — under a quarter of the window area.

Table 1 gives the consequence for one concrete proposal. The reference legend names 17 cover classes, so a confusion matrix over them, at a cell size that can be read, requires on the order of 500 px of width. Seating that in a widened right column at the minimum window leaves roughly 260 px of canvas between the two columns.

Table 1: Width budget at the minimum window of 1000 px.

	As built	With a 500 px matrix column
Left column	240 px	240 px
Right column	255 px	500 px
Canvas	505 px	260 px
Canvas share	50.5 %	26.0 %

The budget is the reason the arrangement cannot be resolved by making surfaces larger. Screen area is fixed; only its allocation over time is free, which is the argument for task-bound layouts in Section 6.3 and against adding permanent chrome.

4 Current surfaces

Table 2 records the studio as built, from a static read of `BoardSurface.tsx` and `MapScreen.tsx`.

Two properties of this arrangement bear on the premise in Section 2.

The detail band is defined only for two planes. It mounts when the selection path contains exactly two predictions. For one plane it does not mount; for three or more, the expression that selects "the two" is not defined by the analysis. The rule that assigns content to surfaces by *arity* — one plane to the column, two to the band — is well defined at $N \leq 2$ and undefined above it.

Both modal dialogs require covering the 3D view. Drawing an area and comparing two rasters are each implemented as a full-screen scrim. The comparison is opened by a gesture performed *on* the board, and its result hides the board that gesture referred to.

5 Blender as a reference architecture

Blender is examined because it is a mature interface for a professionally dense task, in continuous use and revision for approximately three decades, with its design rules stated explicitly rather than inferred.

5.1 Non-overlap as a data-model invariant

Blender's Human Interface Guidelines state the rule directly: the interface should present the relevant options "at a glance, without the need for pushing or dragging windows around," for which reason it defaults to a subdivided window layout. The subdivision operates on three levels: *Screens* (the whole layout), *Areas* (containers), and *Regions* (header, toolbar, sidebar, footer, main region inside an editor).

The mechanism is stronger than the guideline. Each area holds four corner vertices that are shared with its neighbours, so resizing moves a shared edge and neighbouring areas follow by construction. Overlap is not discouraged by convention; it is unrepresentable in the structure. A rule enforced by the data model cannot erode through incremental change, which is the relevant property for a long-lived interface.

Table 2: Reading and input surfaces of the studio. Widths are the values in the source; the two columns and the run band are unconditional.

Surface	Kind	Extent	Carries
`BoardSidebar`	column	15 rem, left	scene tree, data, areas
`BoardStatsBar`	column	15.94 rem, right	one plane: legend, accuracy
`BoardSolarDetail`	band	foot, 1.25–22 rem	two planes: delta, transitions
`BoardRunBar`	band	foot, 4 rem	run parameters (input)
`BoardCompareModal`	modal	covers the view	swipe, domain shift, matrices
`BoardAreaModal`	modal	covers the view	polygon drawing (input)
`boardScene`	canvas	full, beneath	the rasters themselves

5.2 Areas, editors and workspaces

Any area can host any editor type, and the state of every editor that has occupied a given area is retained, so switching type is reversible without loss. Workspaces are named layouts — eleven ship by default, among them Layout, Modeling, Sculpting, Shading, Animation, Compositing and Geometry Nodes — each a saved arrangement of areas and editors bound to a task.

This is the mechanism most relevant to the trade-off that motivated this document. Blender's response to information density is not to display more simultaneously; it is to make the arrangement a property of the task and let the user switch. Tools for sculpting are not on screen while animating, and their absence is not a limitation but the design.

A second mechanism relaxes the tiling temporarily rather than permanently: an area can be maximised to the whole window and restored, and a subset of objects can be isolated in the viewport. Both give one surface the full window without introducing an overlapping layer, so the state remains a property of the partition and is reversible by the same gesture that produced it. This is what the guidelines offer in place of a dialog when a task genuinely needs the whole screen.

5.3 Pinning, and comparison without a dialog

Editors follow the active selection by default, which is what allows one surface to serve many purposes. Pinning breaks that link on demand: a pinned properties editor continues to display the object it was pinned to while the selection moves elsewhere.

The consequence is a general substitute for a comparison dialog. Two editors, each pinned to a different subject, placed side by side by subdivision, compare those subjects without concealing anything and without a device built for the purpose. The mechanism generalises past two — k pinned editors compare k subjects — and it composes with the aggregate view rather than replacing it.

5.4 Context sensitivity

Panels resolve their contents through an ordered chain rather than from global state: the layout context is consulted first, then the region, then the area, then the screen. Consequently one physical surface serves many purposes without configuration — the Properties editor changes its tab set with the active object, and the sidebar shows properties of the active item. Pinning exists as an explicit escape from this coupling when a panel must stop following the selection.

5.5 Devices for density

Recurring mechanisms, applied consistently across editors:

- A scope selector that re-lenses one widget onto different slices of the data model (the Outliner's display modes: View Layer, Scenes, Blender File, Data API, Library Overrides, Unused Data).
- A filter popover cutting on structural and state axes rather than name, with an invert toggle that doubles every state filter.
- Per-row state columns — visibility, selectability, renderability — with modifier keys for subtree propagation, so the state of hundreds of objects is legible in one icon grid.
- Graded expansion operators, from a single row to the whole tree, so row count is a dial rather than a consequence.
- Panels and sub-panels as progressive disclosure, with disclosure state saved in the document rather than reset per session.
- Search that dims non-matching entries and switches tabs, instead of reducing the view to a filtered list.
- An *Operate then Settings* pattern: an operator runs immediately with previous settings and exposes an adjustable region afterwards, replacing the pre-flight parameter dialog.
- The Spreadsheet editor, introduced in version 2.93, presenting per-element data as a filterable table.

6 Correspondence and its limits

6.1 Structural correspondence

The studio arrived independently at an arrangement close to Blender's, which suggests the arrangement is a response to the shape of the problem rather than to a particular tradition.

Table 3: Correspondence between studio surfaces and Blender editors.

Studio	Blender	Correspondence
`boardScene`	3D Viewport	direct
`BoardSidebar`	Outliner	direct; lacks per-row state columns
`BoardStatsBar`	Sidebar (N) / Properties	direct; not context-routed by tabs
`BoardSolarDetail`	horizontal editor at the foot	partial
`BoardRunBar`	tool header / tool settings	direct
`BoardCompareModal`	*none*	no counterpart
`BoardAreaModal`	*none*	no counterpart

The two rows without counterparts are the two modal dialogs. Blender's answer to a comparison task is to subdivide the workspace so both subjects remain visible, which is the same operation the studio performs conceptually when it lifts two rasters onto one surface — and then abandons by opening a dialog over them.

6.2 The studio is one editor, not the container

The correspondence in Table 3 invites a reading that should be stated and rejected: that the studio is Blender's window and its surfaces are Blender's editors, so anything missing must be added *inside* the studio.

Blender does not resolve a demand for tabular data by extending the 3D Viewport's sidebar. It gives an area a different editor type. The correspondence therefore runs one level up: the studio is *an editor*, and the application already has others. `pages/AnalysisPage.tsx` carries the figures, tables and the research pack; `CompareAnalyses.tsx` carries side-by-side comparison including the domain-shift section, phenology series and class statistics.

What the application lacks is not those editors. It is that they are *screens* — navigated between, one at a time, mutually exclusive — rather than *editors* that can occupy areas of one partition. Under Blender's model, examining a raster in the studio while its tables stand open beside it is an ordinary arrangement; here it requires leaving one screen for another.

This reading also disposes of a proposal the analysis initially favoured. Adding a tabular pane to the studio would create a third comparison surface alongside the studio's own and `CompareAnalyses`, with no account of which is authoritative when they disagree. The transfer that follows from the analogy is to make the existing editors tileable, not to grow one of them.

6.3 Mechanisms that transfer

1. **Comparison by subdivision and pinning rather than by dialog.** Removes the contradiction of a comparison hiding its own subjects, and admits $N > 2$ without redefinition: k pinned surfaces compare k subjects. Where a task needs the whole window, maximising an area supplies it without an overlapping layer.
2. **Task-bound layouts.** Classification, comparison and diagnosis place different demands on the same space. Domain-shift diagnosis is a distinct task and need not share a layout with classification.
3. **Context routing of the selection column.** Replaces the arity rule with a mechanism that does not break at $N = 3$: the column shows the active item, and the aggregate view shows the set.
4. **A tabular surface for per-element data.** One row per AOI, sortable by divergence, accuracy, quantity and allocation disagreement. This is the surface Figure 2 selects from, and the natural home for the fields listed in Section 3.2. As Section 3.2 records, the table component and its builders already exist; what is absent is a studio surface that routes to them.

6.4 A precondition the correspondence does not supply

Replacing the comparison dialog with pinned side-by-side surfaces carries a requirement that Blender does not raise, because Blender's editors compare objects and not measured fields.

Two rasters are not comparable by eye unless they share a colour mapping. A continuous layer whose ramp is rescaled to its own range, placed beside another rescaled to its own, will present two different values in the same colour and two equal values in different colours. The reading is confident and wrong, which is a worse failure than the dialog it replaces, since the dialog at least prevented the comparison from being attempted casually.

The mechanism this needs is not in Blender. It is in scientific visualisation: ParaView separates the colour transfer function from the view and exposes explicit rescaling — to the data range, to a custom range, or to the range over all timesteps — so that a shared mapping is a stated choice rather than a side effect. Any side-by-side arrangement in the studio requires an equivalent: one legend domain, reconciled across the panes, with the reconciliation visible.

6.5 Mechanisms that do not transfer

- **General-purpose editors.** Blender's editors are general: a graph editor edits any animatable value. The studio's surfaces are specific to the structure of a run, and a panel that is empty for a single AOI is not an editor in Blender's sense.
- **The learning investment.** Blender's interface is oriented to practitioners who use it for years, and its learning curve is a documented and frequently reported cost. Adopting its flexibility adopts part of that cost. The studio's audience is stated as research-level analysis, which is a comparable profile, but the trade-off is a choice rather than a consequence.

- **Absence of long-running jobs.** A TERRA run is asynchronous, remote and may fail. Blender's closest analogue is rendering, which is handled by a dedicated editor rather than by the general layout mechanism, so the run band has no direct counterpart to inherit from.

7 References outside Blender

Blender answers the layout question well and is silent on several requirements specific to measured data. Three other mature applications supply mechanisms that Blender has no reason to have.

ParaView. Views are first-class occupants of the tiled partition — render view, spreadsheet view, line chart, parallel coordinates — and the same source can be shown in several view types at once with *linked selection* between them, so selecting rows in the spreadsheet highlights the corresponding elements in the 3D view. This is a closer fit than Blender's editor-as-slot for the case in Section 2, where one set of AOIs must be read as a scatter, as a table and as rasters simultaneously. Two further devices apply directly: the colour transfer function is decoupled from the view with explicit rescaling (Section 6.4), and visibility in the pipeline browser is scoped to the active view rather than global, which is what a multi-pane arrangement requires and Blender's single global outliner toggle cannot express. ParaView's Information panel is also a schema-driven readout of the active source's arrays, ranges and element counts, defined over the data model rather than per payload — a low-cost route to a large field count that needs no pane type per product.

QGIS. The Processing History panel records every algorithm invocation with its complete parameter dictionary, re-runnable from the record, alongside batch processing over many inputs. This is the provenance and repeatability device for expensive asynchronous operations, which is TERRA's cost model and is precisely where Blender's *Operate then Settings* does not apply — that pattern is safe only because a Blender operator is cheap to re-run, and a TERRA run is minutes long and consumes quota. QGIS also ships the same content as both a docked panel (Layer Styling) and a dialog (Layer Properties), which is a shipped precedent for the studio's unfinished `placement` distinction.

ImageJ/Fiji. The ROI Manager makes named regions reusable across any open image, with measurements appending rows to one cumulative results table. This makes the region orthogonal to the run: comparing one AOI across N periods becomes N rows in one table rather than N objects on a board, which bears directly on the saved-area facility recently added. Synchronize Windows locks pan, zoom and cursor across open images with a shared crosshair, and is the matured form of the studio's existing link-rasters toggle.

8 Position as a non-geographic encoding

The studio's premise is the removal of cartographic coordinates. That removal frees the coordinate system, and the freed system currently encodes nothing: planes rest where they were dragged.

A candidate use is to encode domain space — placing each AOI at the projection of its feature fingerprint, so that proximity on the board denotes proximity in feature space and an AOI on which the model fails is visibly separated from the others. Under this arrangement the spatial structure of the transferability problem becomes directly observable, which is an operation a cartographic interface cannot perform, since its coordinates are already committed.

Two constraints apply. Projection from a high-dimensional fingerprint to two or three dimensions is lossy, and both PCA and t-SNE produce configurations whose absolute coordinates are not interpretable; relative separation is the only reading the projection supports, and t-SNE in particular does not preserve global distances. Second, if position encodes domain then position is no longer available for manual arrangement, which argues for an explicit arrangement mode — manual, domain, geography, accuracy — rather than a replacement. A mode selector on a view is the same device as the Outliner's display mode.

9 Defects observed while preparing this record

Two are recorded here because they were verified during the audit and are independent of any proposal above.

The domain-shift report is broken at every link. The sidecar defines `mmd_rbf` and emits the key `mmd_rbf`; the Go type declares `MMDLinear` carrying the JSON tag `mmd_linear`; the interface reads `report.mmd_linear`. The name changed at the source and nothing downstream followed, so the figure the panel displays cannot be populated. The sidecar's own test suite calls `ds.mmd_linear`, a function the module no longer defines, so the tests do not pass either. Separately, `standardised` is derived from the standardised moments `z_mean` and `z_var`, which the frontend fingerprint type does not carry.

Escape during drawing closes the studio beneath the dialog. The modal shell binds no keyboard handler; the studio binds a window-level Escape that closes it; and the area-drawing dialog is mounted outside the gate that tracks whether the studio is open. Pressing Escape while drawing therefore dismisses the surface underneath a dialog that remains on screen.

Both are small and neither is a consequence of the architecture discussed here. They are noted so that the audit that found them is not lost with it.

10 Limitations

- The field counts in Section 3.1 come from a static traversal of type declarations. They count declared fields, not fields populated at runtime, and optional members are counted once regardless of how often they appear.
- Figure 2 uses illustrative values. No transferability study has been run in TERRA to date, so the relationship between divergence and accuracy is not measured here and the figure demonstrates an arrangement rather than a result.
- The divergence measure available in the studio is a maximum mean discrepancy with a Gaussian kernel, which is not the $d_{\mathcal{H}\Delta\mathcal{H}}$ of Eq. (1). MMD is used as a computable proxy; the substitution is standard practice and is not justified by the bound itself.
- Statements about Blender, ParaView, QGIS and ImageJ describe documented behaviour and published guidelines. No user study is cited, and the claim that these arrangements work is an argument from adoption and longevity, not from measurement.
- The field counts in Section 3.1 and the budget in Section 3.4 were produced by static reading. The budget assumes the minimum window; at larger windows the canvas share rises and the constraint relaxes, so the figures bound the worst case rather than describing the typical one.
- The correspondence in Table 3 is structural. Structural similarity does not establish that the mechanisms transfer, which is the subject of Section 6.3.

- No interface change described here has been implemented or evaluated. The document records a design position; the position is untested.

11 Summary

The studio exists to support evaluation of a fitted model on regions it was not fitted on. That task has a star topology with one privileged source domain, and it resolves to a single aggregate figure — divergence against accuracy, one point per AOI — that scales linearly in the number of areas examined. The interface as built is organised around pairwise comparison, which is defined at $N \leq 2$ and undefined above it, and it discards three quantities that qualify and localise the distances it does display.

Two constraints bound every response. Screen area is fixed by a minimum window of 1000×700, at which the existing chrome already takes half the width and a third of the height, so no arrangement is reachable by enlarging a surface. And the application already possesses the editors that a first reading of Blender would propose building: what is absent is that they are exclusive screens rather than editors that can share one partition.

The central lesson from Blender is accordingly not that density is solved by displaying more, but that it is managed by binding arrangements to tasks, routing one surface by context, and subdividing instead of opening a dialog — which addresses the studio's one structural contradiction, a comparison that conceals the subjects it compares. Blender is silent on what makes two measured rasters comparable at all; that requirement, a shared and explicit colour mapping, comes from scientific visualisation instead.

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